

PAEDIATRIC



Safe for kids? AI medical devices in radiology overlook paediatric suitability

Brendan S. Kelly^{1*} , Jennifer Lee¹, Edward Antram², Owen Arthurs^{1,3} and Susan C. Shelmerdine^{1,3,4}

Abstract

Objectives Prior publications have highlighted inconsistent labelling of intended use cases and target population for U.S. Food and Drug Administration-approved AI medical devices, especially for children. The extent to which this issue applies to devices within Europe remains unaddressed.

Materials and methods A comprehensive review was conducted of all regulatory-approved AI medical devices for use in radiology from a non-profit, publicly available database. Two independent reviewers assessed information about all the devices regarding use case, modality and intended population. A third reviewer resolved any discrepancies. Where the intended population was unclear, a standardised review of the available evidence and marketing materials for the AI device was conducted.

Results Only four (4/213, 2%) AI medical devices were clearly labelled for paediatric use. A further 11% were intended for all ages, including children. The majority (88/213; 41%) of all AI medical devices did not clearly demonstrate their intended population on the database. Further examination of the scientific literature and marketing of these “unclear” devices showed that 6 (6/88, 7%) of these included patients under 18 in their intended target population, but 47% (41/88) still remained unclear after further review.

Conclusion Most regulated radiology AI medical devices have missing or unclear information regarding the appropriate use in children. This poses significant potential risk, including inadvertent off-label use, which could compromise patient safety. The EU AI Act emphasises the need for transparency and accountability in AI device deployment, and we therefore advocate for a standardised paediatric safety indicator to clearly communicate suitability.

Key Points

Question To what extent do European-approved radiology AI devices provide clear labelling of intended use and suitability for paediatric populations?

Findings Many radiology AI medical devices in Europe lack explicit paediatric use information, raising concerns about unintended off-label use.

Clinical relevance Clear labelling of AI device suitability for children is essential to ensure safe use. A standardised safety indicator could aid clinicians in appropriate device selection.

Keywords Paediatric radiology, Artificial intelligence, Indications for use, Medical devices

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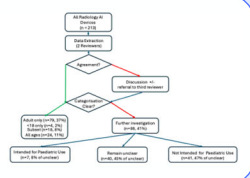
Full list of author information is available at the end of the article

Graphical Abstract

Safe for Kids? AI Medical Devices in Radiology Overlook Paediatric Suitability

Clear information regarding the suitability of intended use of AI medical devices for children is critical to prevent inadvertent off-label application and ensure patient safety.

A comprehensive review was conducted of all regulatory approved AI medical devices for use in radiology from a non-profit, publicly available database.



Paediatrics



AI



Review



Intended Population	Devices	Age (%)
Adult Only	79	37
Paediatric Only	4	2
All Ages	24	11
Subsets of Paediatric Patients	18	8
Unclear	88	41
Total	213	100

Distribution of Marked Radiology AI Devices by Intended Patient Population.

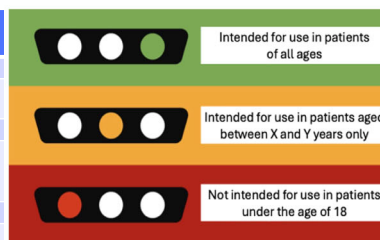


Illustration of a Potential Standardized Paediatric Safety Indicator for Device Labelling

Clear information regarding the suitability of intended use of AI medical devices for children is critical to prevent inadvertent off-label application and ensure patient safety. Implementing a standardised safety indicator could guide clinicians in the appropriate selection and use of AI tools.

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Introduction

AI as a Medical Device (AIaMD) is defined as “the use of artificial intelligence (AI) Technologies in medical applications with the intention of diagnosing, treating, monitoring or preventing diseases” [1, 2]. As with any medical device, AIaMDs require strict regulatory and conformity assessments prior to clinical implementation. One of the key features of the assessment should include a clear statement of the intended use of the device. This should highlight information on the specific disease being targeted, the imaging modality it works for (if radiology-based), the population evaluated, as well as the intended target population [2, 3]. Recently, greater awareness of AI ethical practices and bias have raised concerns regarding the applicability and safety of AI devices across diverse patient populations, particularly paediatric patients [3, 4]. Paediatric radiology presents unique challenges for the development of AIaMD due to the distinct anatomical, physiological, and developmental characteristics of children [5–7]. Their continual growth, varying disease presentations, and differences in anatomical maturity necessitate tailored diagnostic approaches [6, 8]. AI algorithms developed and validated primarily on adult datasets may also not perform adequately when applied to paediatric cases, potentially leading to diagnostic inaccuracies or missed findings [4].

Inconsistent intended use statements for AIaMD, particularly relating to their intended target patient populations, exacerbates this issue [4, 9]. In the United States, for instance, AI devices cleared by the Food and Drug Administration (FDA) often have general indications for use, making it challenging to determine their applicability to paediatric patients [4], and it has been shown that up to 95% of intended use statements are inadequate [9]. When the lack of clear intended uses exists, there is uncertainty amongst clinical users regarding appropriate use. This ambiguity is particularly problematic in hospitals serving both adults and children, where the risk of inadvertent off-label use increases, potentially compromising patient safety and leading to regulatory non-compliance [1, 9, 10].

The recent 2024 European Union (EU) AI Act introduces stringent requirements for AI systems, emphasising transparency, accountability, and rigorous validation [11–13]. These provisions are particularly relevant to medical devices used in vulnerable populations, like as children. The Act presents an opportunity to address the existing gaps in labelling and validation of AI devices for paediatric use [3]. This study, therefore, aims to evaluate the extent to which radiology AI devices include paediatric patients in their intended use.

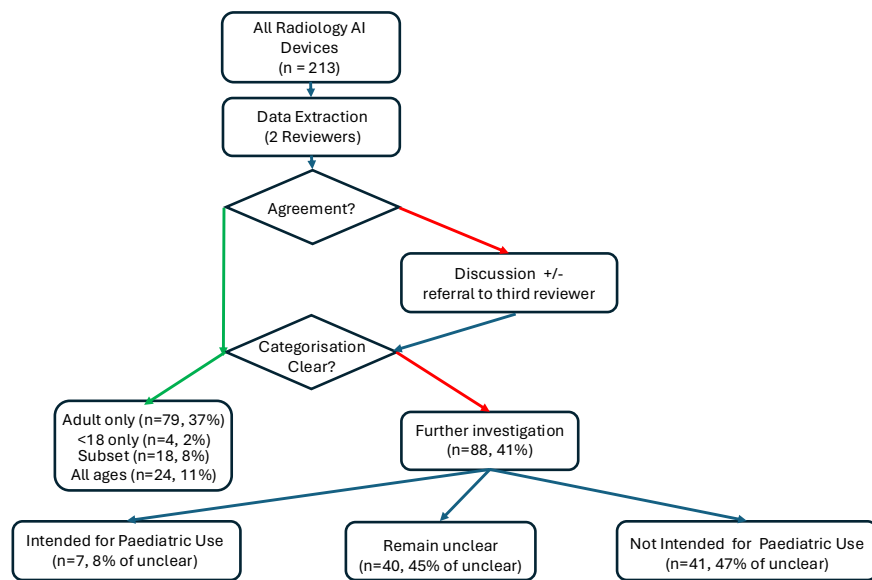


Fig. 1 Flowchart of device review and categorisation process. Blue arrow—signifies default flow of information, and green arrow—signifies flow if yes, red arrow if no

Materials and methods

Ethical approval

Ethical approval was not required for this review of publicly available information. No patient identifiable information was included in this review. No hospital or personal data was accessed to perform our analyses.

Data source

A comprehensive retrospective review of all listed radiology AI devices with regulatory approval was conducted using the database radiology.healtharegister.com (Fig. 1) [14].

This database has been active for the past 3 years and is currently provided by an independent organisation (Romion Health, Utrecht, Netherlands; <https://www.romionhealth.com/>). The database is the most comprehensive available listing of all AI-based medical devices intended for clinical radiology practice. The database is free to access, and vendors are not required to pay to be displayed on this website, making it an unbiased resource.

The website was scraped using the python code provided (Supplementary Data 1), and a csv file was compiled for analysis (Supplementary Data 2). The data extraction was performed on 04/08/2024 and updated again on 31/12/2024 prior to manuscript submission to ensure all data was contemporaneous.

As a secondary measure, we also cross-checked all the AI devices found on radiology.healtharegister.com with another device database (HARI—Hardian Regulatory Intelligence [15]), beta launched more recently in September 2024, containing similar information on

available clinical evidence, reported adverse events and intended uses of all AIaMD products, not just limited to radiology.

Data analysis

All available AI tools listed on the publicly available database were evaluated. These included medical devices intended for diagnostic or analytical purposes in radiology using AI. We did not include any devices that were not relevant to the field of radiology.

Two independent reviewers (senior fellows in paediatric radiology with expertise in radiology and medical device regulation—both with 2 years post radiology fellowship examination) extracted data on each device's intended use, specified patient population, and any paediatric indications. A third reviewer (post-doctoral academic, 10 years paediatric radiology experience) was available to resolve any disagreements to ensure accuracy and consistency.

Devices were categorised based on their intended patient population into five groups:

- Adult only,
- Paediatric only,
- All ages (adult and paediatric),
- Subsets of paediatric patients (e.g. prespecified age group), and
- Unclear.

The “Unclear” category included devices with broad intended uses or insufficient information regarding patient age.

Where the devices demonstrate unclear labelling, further investigation was conducted by looking at the vendor's websites, publicly available product documentation (e.g. FDA 501k) and available scientific literature. Where this information was challenging to find, we reviewed the Medline database for peer-reviewed articles where the AI product was used to look for additional information about the AI product and intended use statements, if included. Supplementary Data 3 outlines this decision process.

A narrative review and simple statistics (e.g., percentages and total frequency of occurrences) were reported.

Results

Device information

Both reviewers agreed on the data extraction findings of devices for all products, without the need for intervention from the third reviewer. The raw data detailing the individual AIaMDs for radiology, their vendors with intended use statements and the indicative target population are available in the supplementary material.

An initial search of the public database revealed a total of 213 AIaMD products relevant to the field of radiology (Table 1). Of these, 4/213 (2%) were categorised as “paediatric only”, 18/213 (8%) for “subsets of paediatric patients”, and 24/213 (11%) as applicable to ‘all ages’ (indicating both adult and paediatric population).

In 88/213 (41%), it was unclear from the public database who the intended target population was. Of the AIaMD tools for radiology, the most common modality with unclear labelling was CT (43/88, 49%), with neuroradiology (42/88, 48%) and stroke imaging (13/88, 15%) representing the most common radiological subspecialty and pathological process (Figs. 2–4). Unclear labelling was also seen across a wide variety of vendors (Fig. 5).

Upon further evaluation of the vendor websites, marketing materials, public regulatory documentation and scientific literature, it was found that of these 88 AIaMDs (40/88, 45%) were intended for use in adults only, and 6 (6/88, 7%) were intended for children under 18 only. This included devices intended for use in patients over 15 years in screening for tuberculosis with chest radiography. The

Table 1 Distribution of marked radiology AI devices by intended patient population

Intended patient population	Number of devices	Percentage (%)
Adult only	79	37
Paediatric only	4	2
All ages	24	11
Subsets of paediatric patients	18	8
Unclear	88	41
Total	213	100

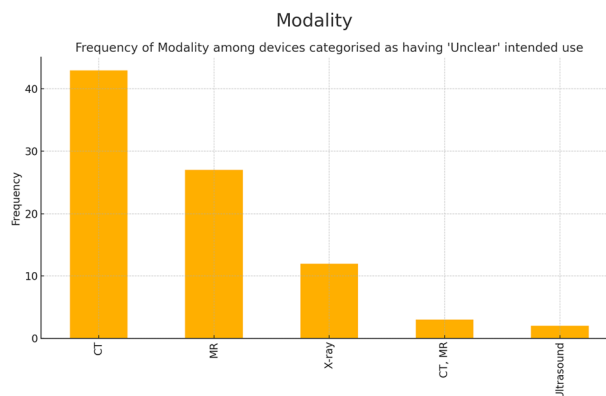


Fig. 2 Frequency of different imaging modalities used in the 88 devices categorised as “unclear” based on the labelling of intended use on radiology.healthregister.com

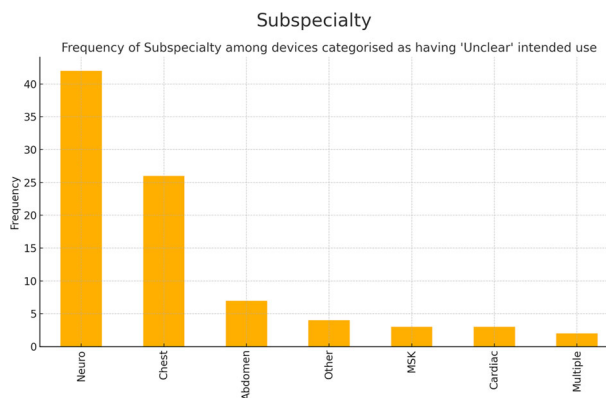


Fig. 3 Frequency of different radiologic subspecialties used in the 88 devices categorised as “unclear” based on the labelling of intended use on radiology.healthregister.com

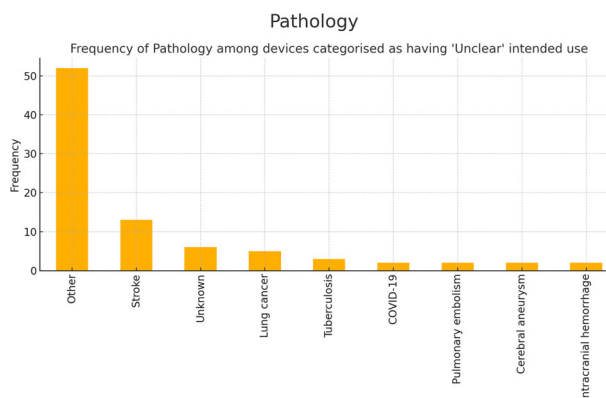


Fig. 4 Frequency of different pathologies used in the 88 devices categorised as “unclear” based on the labelling of intended use on radiology.healthregister.com

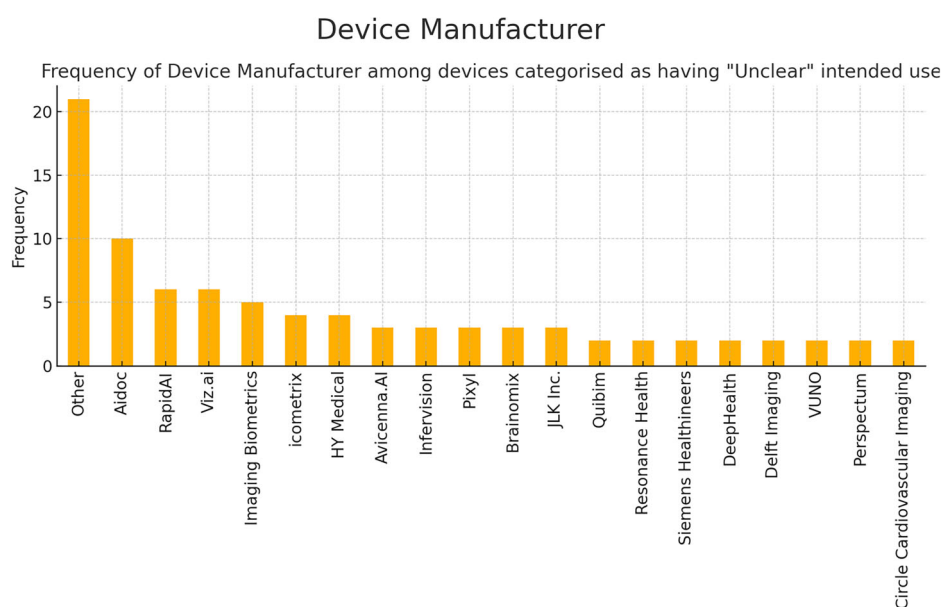


Fig. 5 Frequency of different AIaMD manufacturers in the 88 devices categorised as “unclear” based on labelling of intended use on [radiology.healthregister.com](https://www.healthregister.com)

remainder (41/88, 47%) were still unclear after further investigation.

Overall, a total of 11/213 AIaMD in our study were intended for use by children only, but only 4 made this explicit without further exploration.

Discussion

This study has demonstrated that a considerable proportion of commercially available Conformité Européenne regulated AIaMD products for radiology lack information regarding whether they are intended for use in children. It also found that there is a lack of paediatric appropriate AIaMD tools for use in radiology in general, highlighting the need for more innovation and regard for this underserved population. These findings are highly relevant, as there are risks of using AI-enabled software inadvertently in a population it is not intended for, leading to adverse events in clinical practice.

The high percentage of AIaMD devices with unclear labelling suggests that manufacturers are either unaware of the need for this clarity, may not prioritise paediatric validation or specific regulation for this population, or may assume that devices applicable to adults are equally suitable for children. Even those tools labelled and described as being suitable for ‘all ages’ appear more relevant to diseases that predominantly affect adults (e.g., dementia, coronary artery disease), and thus it is unclear if this was genuinely for ‘all ages’ or inaccurate wording actually intended to be ‘all adults’. The reasons for poor labelling and paediatric AI tools development may relate

to the difficulty in realising a return on investment for the additional evidence to prove a tool is valid for use in children or the potential for liability if adverse events occur in children. Regardless of the underlying causes, increased awareness of this problem and providing better clarity is vital so that vendors and consumers (both healthcare professionals and patients/carers) can make better informed decisions.

Our findings align with concerns raised in previous studies where over 40% of ‘intended use’ statements for AI tools were found to be incomplete [9], making it challenging to know exactly what clinical problem the AI tool intends to solve. In an analysis of FDA-approved AI tools as medical devices in the United States, many were identified as having only general indications for use [4]. A more recent study [16], looking at all FDA-approved AI tools for children’s healthcare (not necessarily specific to radiology), found that only 54/149 (36%) included information on patient ages in their validation datasets, with the vast majority being unknown (99/149, 66%) or having performed the validation dataset on adults, 22/149 (15%), even though the tool was authorised for children’s use. Together, this demonstrates the challenges faced by healthcare providers when trying to make informed decisions regarding the suitability of AI tools for children.

Our study, combined with work from previous authors, confirms the fact that the lack of information and transparency in how AI tools should be used is a global problem, and highlights the urgent need for improved regulatory guidelines and manufacturer practices

regarding specific labelling, especially whether the intended use and target populations include children. Knowledge of this gap now allows us the opportunity to improve our understanding of the landscape for AI in paediatric radiology [17] and try to meet the expectations of our stakeholders [18]. We echo previous calls to action from healthcare providers [4] to directly contact the device vendor and explicitly clarify whether information on the performance of the device in paediatric patients is available before purchasing or using such devices in the care of children, and for vendors to make it explicit whether this is the case or not in their marketing and regulatory materials.

In our analysis, we have noticed some interesting trends amongst the AIaMDs with unclear labelling. The commonest AIaMD with unclear labelling was neuroimaging, especially for the detection of acute ischaemic stroke using CT. This could be explained by the fact that this was an early flagship use case for AI in radiology [14] and many early regulatory approvals were granted in this field before there was more awareness of the need to establish transparency and lack of bias [19]. Childhood stroke, while less common than in adults, remains an important cause of neurological morbidity and mortality [20], but has a different pathophysiology, which could affect the accuracy of any previously approved AI tools for this purpose. Furthermore, intracranial haemorrhage is also commonly investigated with head CT and is seen in children with traumatic [21] and non-traumatic [22] aetiology, sometimes as a marker of underlying physical abuse and requiring urgent safeguarding decisions. Reliance on off-label use of AIaMD for these time-dependent and serious conditions, with potential medicolegal implications, may not be appropriate in such settings where the accuracy of the tools is unknown and regulatory approval is not provided.

To mitigate these risks, careful evaluation and understanding of AI tools are required prior to implementation [23]. AI tools intended for use in children should have specific labelling and safety ratings for paediatric use, as advocated by the ACR Image Intelligently Campaign [24] and suggestions for data health equity for children [7]. A standardised paediatric safety indicator (such as a traffic light system) can guide healthcare providers in understanding which tools require more critical appraisal and oversight at a glance: a simplified illustrative template is provided in Fig. 6. For example a device such as “Myrian XP-Prostate option AI” whose intended use refers to “Male adults” would have a red light assigned in a paediatric safety indicator. A device such as “BoneMRI”, whose intended use clearly refers to “Patients from 12 years and older suspected of having conditions involving the bony anatomy of the lumbar spine, cervical spine or

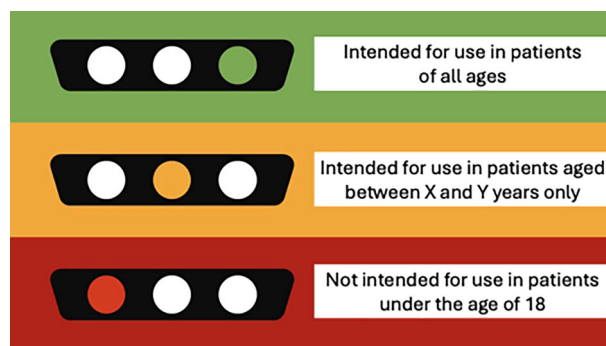


Fig. 6 Illustration of a potential standardised paediatric safety indicator for device labelling

pelvic region” would receive a yellow indicator. Only devices explicitly intended for use in all children, such as “BoneXpert”, would receive a green indicator.

The proposal of integrating a standardised paediatric safety indicator into device labelling offers a practical solution to enhance clarity and safety. There could be an additional ‘quick view’ measure on top of “model cards” that have been advocated by Mitchell et al [25] and supported by the OECD [26]. This could provide benchmark accuracy of the AI model in a variety of conditions, such as across different cultural or demographic groups that are relevant to the intended use, and may prevent inadvertent use on non-approved populations in a wider healthcare setting.

This work fits into a larger concern around children being left behind in AI development in healthcare. Comprehensive training datasets across all paediatric subgroups are currently lacking, including appropriate diversity and representation, as suggested by ACCEPT—guidelines [27]. Transparency and explainability are key when using clinical decision-making tools in under-represented populations like children [3, 18], and help build trust [9, 18, 23]. Recent regulatory developments, including the FDA’s latest draft guidance for AI/ML devices [28], reinforce the need for rigorous performance validation, transparent labelling, and demonstrating predictable and reliable outcomes in the intended well-defined target populations. We believe that companies/device manufacturers that embrace these principles could benefit from increased trust from potential customers and that the integration of such safety indicators into policy recommendations and regulatory incentives in the European context could increase trust in AIaMD overall. Trust and transparency are key to the successful adoption of AI, and in line with how the EU Paediatric Regulation (EC No 1901/2006) sets up a system of requirements, rewards and incentives, to ensure that medicinal products are researched, developed and authorised to meet the

therapeutic needs of children, the same will be necessary to ensure children are not left behind. Further work is needed, however, to outline a roadmap for how this could be achieved in practice.

This study has several limitations. The analysis was limited to devices listed on one database (radiology.healthcareregister.com), which is ever evolving as new AI tools become available on the market. Furthermore, some medical devices may not be included in this database, such as those that are not directly relevant to radiology, but may rely on medical photography and carry similar risks for use in children (if only trained on an adult dataset). We tried to mitigate against this by performing our search criteria twice. Furthermore, the databases used may not encompass all globally available AI devices, potentially introducing selection bias.

Secondly, reliance on publicly available information and external sources for intended use and population performance metrics may have resulted in incomplete assessments, as manufacturers may have unpublished data regarding paediatric validation that is not openly shared, and similarly those that include labelling as ‘all ages’ may not have been externally validated in children or the implication may have been ‘all adults’ without the vendor being specific (given some of the use cases do not appear to be directly relevant to children e.g. dementia and coronary artery disease). We took the available data in good faith, as it was presented, as this represents what the average consumer would have to hand themselves when deciding on product adoption. We also tried to mitigate for this by cross-checking information provided by one public database with a second, and performing searches of the manufacturer’s website and any published material that might present more comprehensive details of model development.

Furthermore, whilst we provide statistics on unclear labelling, the reasons behind this unclear documentation were not explored, and further work in this field could delve into what barriers are preventing vendors from sharing this knowledge or labelling to get to the root of the issue.

Conclusion

This study reveals that a substantial proportion of regulated AIaMD tools for radiology lack clarity regarding their intended use for children. This poses significant potential risks for patient safety. Our findings underscore the need for AI vendors to provide clear instructions and for regulatory bodies to enforce standards specifying intended patient populations. Integrating a standardised paediatric safety indicator into device labelling and model cards offers a practical solution to enhance clarity and promote safer clinical practices. Collaborative efforts

among manufacturers, regulators, healthcare institutions, and clinicians are essential to ensure AI devices are safe and effective for paediatric patients.

Abbreviations

AI	Artificial intelligence
AIaMD	AI as a medical device
EU	European Union
FDA	Food and Drug Administration

Supplementary information

The online version contains supplementary material available at <https://doi.org/10.1007/s00330-025-11975-6>.

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Compliance with ethical standards

Guarantor

The scientific guarantor of this publication is B.S.K.

Conflict of interest

B.S.K. is an NDTP Dr Richard Steevens and ESOR Fellow in AI and Paediatric Radiology. He is also supported by the Dean Medal for research by MS Ireland. He is the Social Media Editor for *European Radiology* and, as such, did not participate in the selection or review processes for this article. S.C.S. is supported by an NIHR Advanced Fellowship Award (NIHR-301322). S.C.S. declares non-financial research collaborations with Gleamer and Milvue. S.C.S. has previously received honoraria for keynote speaking from Bayer Plc. The remaining authors of this manuscript declare no relationships with any companies, whose products or services may be related to the subject matter of the article.

Statistics and biometry

No complex statistical methods were necessary for this paper.

Informed consent

Written informed consent was not required for this study because it was a review of publicly available data.

Ethical approval

Institutional Review Board approval was not required because it was a review of publicly available data.

Study subjects or cohorts overlap

NA.

Methodology

- Methodology
- Review of publicly available data

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